



# Veliparib with carboplatin and paclitaxel in *BRCA*-mutated advanced breast cancer (BROCADE3): Final overall survival results from a randomized phase 3 trial<sup>☆</sup>

Véronique Diéras<sup>a,b,\*</sup>, Hyo S. Han<sup>c</sup>, Hans Wildiers<sup>d</sup>, Michael Friedlander<sup>e,f</sup>, Jean-Pierre Ayoub<sup>g</sup>, Shannon L. Puhalla<sup>h</sup>, Delphine Loirat<sup>a</sup>, Christine Ratajczak<sup>i</sup>, Hephzibah Adamu<sup>i</sup>, Vincent Girardi<sup>i</sup>, Banu K. Arun<sup>j</sup>

<sup>a</sup> Medical Oncology Department, Institut Curie, 26 rue d'Ulm, 75005, Paris, France

<sup>b</sup> Medical Oncology Department, Centre Eugène Marquis, Av. de la Bataille Flandres-Dunkerque CS 44229, 35000 Rennes, France

<sup>c</sup> Moffitt Cancer Center, 12920 McKinley Drive, Tampa, FL 33612, USA

<sup>d</sup> Department of General Medical Oncology, University Hospitals Leuven, Herestraat 49, O&N4 box 818, 3000 Leuven, Belgium

<sup>e</sup> School of Clinical Medicine, UNSW Sydney, Cnr High St & Botany St., Sydney NSW 2052, Australia

<sup>f</sup> Department of Medical Oncology, The Prince of Wales Hospital, Barker Street, Randwick, NSW 2031 Australia

<sup>g</sup> Centre Hospitalier de l'Université de Montréal, 1000 rue Saint-Denis, Montréal, QC H2X 0C1, Canada

<sup>h</sup> UPMC Hillman Cancer Center, 9100 Babcock Blvd Ste G600, Pittsburgh, PA 15237, USA

<sup>i</sup> AbbVie, 1 N Waukegan Rd, North Chicago, IL 60044, USA

<sup>j</sup> The University of Texas MD Anderson Cancer Center, 1155 Pressler St, Houston, TX 77030, USA

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## ABSTRACT

**Background:** In the BROCADE3 study, the addition of veliparib to carboplatin plus paclitaxel resulted in a significant improvement in progression-free survival (PFS) compared with placebo plus carboplatin and paclitaxel, in patients with germline *BRCA1* or *BRCA2* (*BRCA1/2*)-mutated, human epidermal growth factor receptor 2 (HER2)-negative advanced breast cancer. We now report final overall survival (OS) data.

**Methods:** BROCADE3 is a randomized phase 3 study that enrolled patients with *BRCA1/2*-mutated, HER2-negative advanced breast cancer who received  $\leq 2$  prior lines of chemotherapy for metastatic disease. Patients were randomized 2:1 to carboplatin and paclitaxel, dosed with either veliparib or matching placebo. OS was a secondary endpoint.

**Results:** In the intention-to-treat population ( $N = 509$ ), 337 patients were randomized to receive veliparib and 172 to placebo. Median OS was 32.4 months vs 28.2 months (hazard ratio, 0.916; 95% CI, 0.736–1.140;  $P = .434$ ). The updated safety data for veliparib are consistent with those reported in the primary analysis; the addition of veliparib was generally well tolerated.

**Conclusions:** Final OS data indicate that the PFS improvement shown in the primary analysis did not translate into an OS benefit. The long survival times observed in both arms suggest that combination therapy with paclitaxel and carboplatin provides clinical benefit in the population of patients with *BRCA1/2*-mutated metastatic breast cancer.

## 1. Introduction

Around 5% of breast cancer cases result from inherited mutations in *BRCA1* and *BRCA2* genes [1,2]. These cancers are highly susceptible to cytotoxicity induced by poly(ADP-ribose) polymerase (PARP) inhibitors

and DNA-damaging platinum agents [3–5], although several mechanisms cause resistance to these drugs [6]. BROCADE3 evaluated whether a combination of these agents could maximize potential clinical benefit prior to developing cross-resistance.

Veliparib, a potent, orally bioavailable, selective PARP-1 and PARP-2

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\* Corresponding author at: Medical Oncology Department, Institut Curie, 26 rue d'Ulm, 75005, Paris, France.

E-mail address: [v.dieras@rennes.unicaner.fr](mailto:v.dieras@rennes.unicaner.fr) (V. Diéras).

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inhibitor, can enhance the efficacy of platinum chemotherapy [7]. The phase 3 BROCADE3 trial (NCT02163694) evaluated the addition of veliparib to carboplatin and paclitaxel in patients with germline *BRCA*-mutated, human epidermal growth factor receptor 2 (HER2)-negative advanced breast cancer [8]. Adding veliparib resulted in significant improvement in progression-free survival (PFS) vs placebo plus carboplatin and paclitaxel: median PFS was 14.5 months vs 12.6 months, respectively; hazard ratio [HR], 0.71;  $P = .0016$ . At 2 and 3 years, more patients were alive and progression free in the veliparib vs placebo group (34% vs 20%, and 26% vs 11%). Overall survival (OS) data were not mature at the time of the primary PFS analysis; the pre-planned interim analysis showed a difference of 5.3 months in median OS between the veliparib and control arms (33.5 months and 28.2 months, respectively [HR, 0.95; 95% CI, 0.73–1.23]). The OS analysis, planned when 357 events [deaths] occurred, and updated safety data are reported.

## 2. Methods

Trial design, eligibility criteria, efficacy and safety assessments, and statistical analysis have been published previously [8].

### 2.1. Study design and participants

This randomized, placebo-controlled study compared the efficacy of veliparib vs placebo (control group) plus carboplatin and paclitaxel in patients with germline *BRCA1* or *BRCA2* mutation-associated advanced HER2-negative breast cancer who had received  $\leq 2$  prior lines of chemotherapy for metastatic disease. Patients received veliparib (120 mg orally twice daily [BID]) or placebo on days –2 to 5, carboplatin (area under the concentration-time curve 6 mg/mL per min intravenously) on day 1, and weekly paclitaxel (80 mg/m<sup>2</sup> intravenously) on days 1, 8, and 15 (of 21-day cycles) until disease progression/unacceptable toxicity. Patients discontinuing both carboplatin and paclitaxel (except for disease progression) received continuous single-agent veliparib or placebo at 300 mg BID, increasing to 400 mg BID, if tolerated. Control group patients could receive crossover open-label veliparib monotherapy after disease progression.

The primary endpoint was investigator-assessed PFS according to Response Evaluation Criteria in Solid Tumors version 1.1 [9]; secondary endpoints included OS.

The study was approved by the institutional review board at each site; all patients provided written consent.

## 3. Results

### 3.1. Study population

In total, 513 patients were randomized; 509 patients (veliparib group,  $n = 337$ ; control group,  $n = 172$ ) had a confirmed germline *BRCA1* or *BRCA2* mutation. Among randomized patients, 507 patients received at least 1 dose of veliparib or placebo (veliparib group,  $n = 336$ ; control group,  $n = 171$ ). Baseline demographics and clinical characteristics are presented in Table 1; 81% of patients were chemotherapy naïve.

At final OS analysis (January 4, 2022), 87.8% of patients had discontinued treatment in the veliparib group and 94.2% in the placebo group; disease progression was the primary reason in 61.4% and 71.5% of patients, respectively. Median blinded period exposure to study drug was 117.5 days (min, 2 days; max, 2197 days) in the veliparib group and 115 days (min, 2 days; max, 2067 days) in the control group.

### 3.2. Efficacy

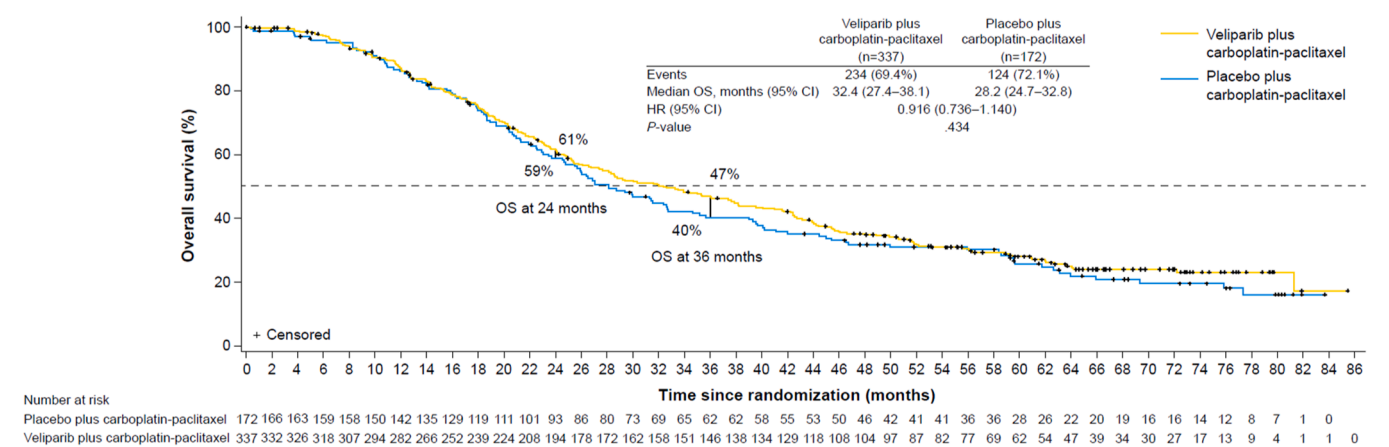
For this final analysis, median follow-up was 65.7 months (95% CI, 62.4–68.8) in the veliparib group and 68.4 months (95% CI, 59.6–76.0)

**Table 1**  
Baseline characteristics in the intention-to-treat population.

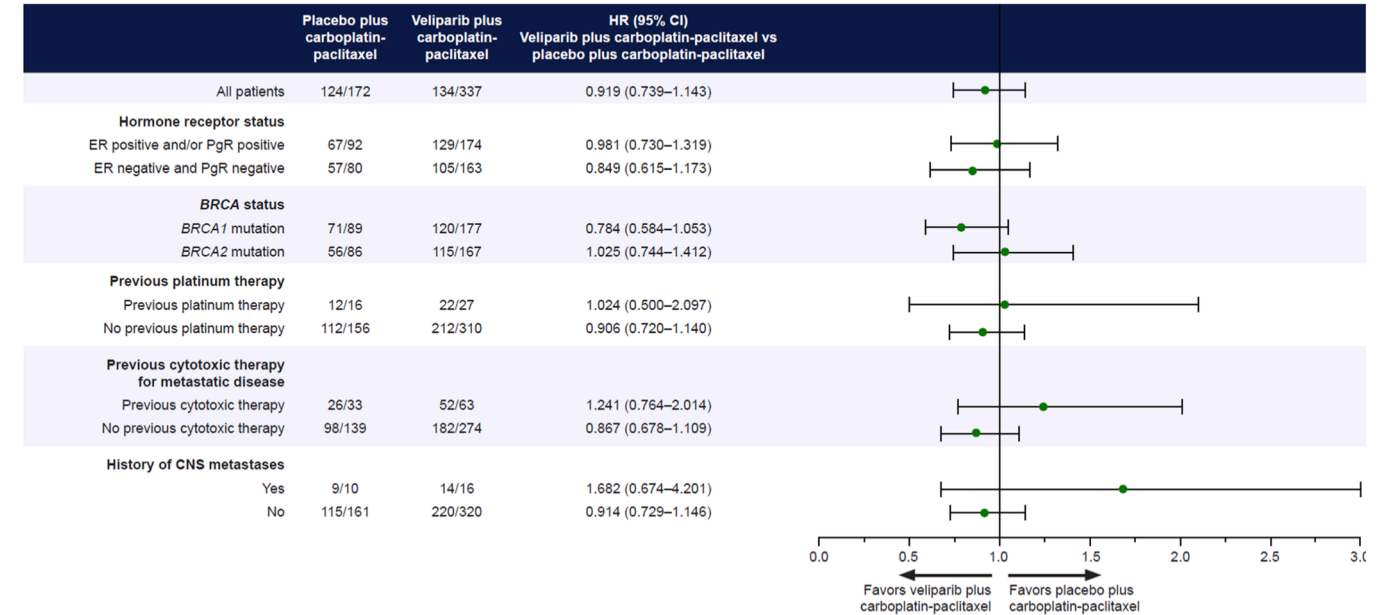
|                                                                          | Veliparib plus<br>carboplatin-paclitaxel<br>( $n = 337$ ) | Placebo plus<br>carboplatin-<br>paclitaxel<br>( $n = 172$ ) |
|--------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|
| Sex, $n$ (%)                                                             |                                                           |                                                             |
| Female                                                                   | 333 (99)                                                  | 169 (98)                                                    |
| Male                                                                     | 4 (1)                                                     | 3 (2)                                                       |
| Age, years (range)                                                       | 47 (39–54)                                                | 45 (39–54)                                                  |
| Race, $n$ (%)                                                            |                                                           |                                                             |
| White                                                                    | 294 (87)                                                  | 153 (89)                                                    |
| Asian                                                                    | 24 (7)                                                    | 12 (7)                                                      |
| Black                                                                    | 14 (4)                                                    | 6 (3)                                                       |
| Other                                                                    | 5 (1)                                                     | 1 (1)                                                       |
| Geographic region, $n$ (%)                                               |                                                           |                                                             |
| USA                                                                      | 46 (14)                                                   | 27 (16)                                                     |
| Other                                                                    | 291 (86)                                                  | 145 (84)                                                    |
| ECOG performance status, $n$ (%)                                         |                                                           |                                                             |
| 0                                                                        | 208 (62)                                                  | 102 (59)                                                    |
| 1                                                                        | 121 (36)                                                  | 63 (37)                                                     |
| 2                                                                        | 8 (2)                                                     | 7 (4)                                                       |
| Measurable disease, $n$ (%)                                              | 285 (85)                                                  | 143/171 (84)                                                |
| Liver or lung metastases, $n$ (%)                                        | 219 (65)                                                  | 104/171 (61)                                                |
| Bone-only disease, $n$ (%)                                               | 11 (3)                                                    | 9/171 (5)                                                   |
| Previous platinum therapy use, $n$ (%)                                   | 27 (8)                                                    | 16 (9)                                                      |
| Previous chemotherapy in neoadjuvant or adjuvant setting, $n$ (%)        | 236 (70)                                                  | 113 (66)                                                    |
| Previous chemotherapy in metastatic setting, $n$ (%)                     | 63 (19)                                                   | 33 (19)                                                     |
| History of CNS metastases, $n$ (%)                                       | 16/336 (5)                                                | 10/171 (6)                                                  |
| Any previous therapy in the metastatic setting, $n$ (%)                  | 90 (26.7)                                                 | 50 (29.1)                                                   |
| Hormone receptor expression, $n$ (%)                                     |                                                           |                                                             |
| Estrogen receptor or progesterone receptor positive                      | 174 (52)                                                  | 92 (53)                                                     |
| Estrogen receptor or progesterone receptor negative                      | 163 (48)                                                  | 80 (47)                                                     |
| <i>BRCA1</i> or <i>BRCA2</i> mutation status by core laboratory, $n$ (%) |                                                           |                                                             |
| <i>BRCA1</i> mutation positive                                           | 177 (53)                                                  | 89 (52)                                                     |
| <i>BRCA2</i> mutation positive                                           | 167 (50)                                                  | 86 (50)                                                     |
| Stage at diagnosis, $n$ (%)                                              |                                                           |                                                             |
| 0                                                                        | 3/332 (1)                                                 | 1/169 (1)                                                   |
| I                                                                        | 43/332 (13)                                               | 22/169 (13)                                                 |
| II                                                                       | 128/332 (39)                                              | 69/169 (41)                                                 |
| III                                                                      | 84/332 (25)                                               | 39/169 (23)                                                 |
| IV                                                                       | 74/332 (22)                                               | 38/169 (22)                                                 |

Abbreviations: CNS, central nervous system; ECOG, Eastern Cooperative Oncology Group.

in the control group. There were 234 events (deaths) in the veliparib group and 124 in the control group; median OS was 32.4 months vs 28.2 months (HR, 0.916; 95% CI, 0.736–1.140;  $P = .434$ ; Figure 1). OS at 24 months was 61.4% (95% CI, 55.8–66.5%) in the veliparib group and 58.9% in the control group (95% CI, 50.9–66.0%); at 36 months, OS was 47.0% (95% CI, 41.1–52.4%) and 40.3% (95% CI: 32.7–47.8%). No significant difference in OS was seen in patient subgroups between the veliparib and control groups (Figure 2). Overall, 82 (48%) patients in the control group had received open-label veliparib (crossover) as first subsequent therapy.



**Fig. 1.** Kaplan-Meier curve for overall survival (ITT population). Placebo plus carboplatin-paclitaxel = placebo 120 mg BID plus carboplatin-paclitaxel; veliparib plus carboplatin-paclitaxel = veliparib 120 mg BID plus carboplatin-paclitaxel. Stratified by prior platinum therapy (yes vs no) and receptor status (ER and/or PgR positive vs ER/PgR negative). BID, twice daily; ER, estrogen receptor; ITT, intention-to-treat; OS, overall survival; PgR, progesterone.



**Fig. 2.** Analysis of overall survival by subgroup (ITT population). Placebo plus carboplatin-paclitaxel = placebo 120 mg BID plus carboplatin-paclitaxel; veliparib plus carboplatin-paclitaxel = veliparib 120 mg BID plus carboplatin-paclitaxel. Tests on subgroups other than ER/PgR status. The ER/PgR subgroup are unstratified. HR < 1 indicates favor of veliparib 120 mg BID plus carboplatin-paclitaxel. BID, twice daily; CNS, central nervous system; ER, estrogen receptor; HR, hazard ratio; ITT, intention-to-treat; PgR, progesterone.

3.3. Safety

In total, 334 (99%) patients in the veliparib arm and 171 (100%) patients in the control arm experienced a treatment-emergent adverse event (TEAE; Table 2). Neutropenia was the most common grade  $\geq 3$  TEAE, reported in 81% of patients in the veliparib group and in 84% in the control group, followed by anemia (42% and 40%) and thrombocytopenia (40% and 28%). There was no significant difference in AEs leading to study drug discontinuation (19.0% vs 11.7% in the veliparib and control groups, respectively;  $P = .043$ ).

AEs leading to death occurred in 8 (2.4%) patients in the veliparib group (malignant neoplasm progression [ $n = 6$ ], pulmonary embolism [ $n = 1$ ], and sepsis [ $n = 1$ ]) and in 3 (1.8%) patients in the control group (malignant neoplasm progression [ $n = 2$ ] and pulmonary artery thrombosis [ $n = 1$ ]). The AEs were considered unrelated to veliparib.

Among 336 patients in the veliparib plus carboplatin-paclitaxel arm, there was 1 patient with treatment-emergent myelodysplastic syndrome

(MDS). Five individual patients had MDS and/or acute myeloid leukemia (AML) occurring > 30 days after the last dose of study drug, with MDS reported for 4 patients and AML for 3 patients. Among 171 patients in the placebo plus carboplatin-paclitaxel arm, MDS occurring > 30 days after the last dose of study drug was reported for 1 patient and there were no reported cases of AML.

4. Conclusions

The primary analysis of the phase 3 BROCADE3 trial, which enrolled > 500 patients with metastatic BRCA-mutated breast cancer, showed a significant improvement in median PFS and a numeric increase in proportion of patients progression free at 2 and 3 years when veliparib was added to the carboplatin-paclitaxel doublet and continued as monotherapy if carboplatin or paclitaxel were discontinued before disease progression [8]. In this final analysis, OS was numerically but not significantly improved by the addition of veliparib to carboplatin and

**Table 2**

Treatment-emergent adverse events occurring in at least 20% of patients.

|                                  | Veliparib plus<br>carboplatin-paclitaxel<br>(n = 336) |                     | Placebo plus<br>carboplatin-paclitaxel<br>(n = 171) |                     |
|----------------------------------|-------------------------------------------------------|---------------------|-----------------------------------------------------|---------------------|
|                                  | All grade,<br>n (%)                                   | Grade ≥ 3,<br>n (%) | All grade,<br>n (%)                                 | Grade ≥ 3,<br>n (%) |
| Any adverse event                | 334 (99)                                              | 325 (97)            | 171 (100)                                           | 163 (95)            |
| Neutropenia                      | 300 (89)                                              | 272 (81)            | 156 (91)                                            | 143 (84)            |
| Thrombocytopenia                 | 271 (81)                                              | 134 (40)            | 122 (71)                                            | 48 (28)             |
| Anemia                           | 271 (81)                                              | 142 (42)            | 119 (70)                                            | 68 (40)             |
| Leukopenia                       | 135 (40)                                              | 98 (29)             | 65 (38)                                             | 47 (28)             |
| Nausea                           | 246 (73)                                              | 22 (7)              | 109 (64)                                            | 7 (4)               |
| Alopecia                         | 182 (54)                                              | -                   | 88 (52)                                             | -                   |
| Fatigue                          | 170 (51)                                              | 24 (7)              | 87 (51)                                             | 7 (4)               |
| Peripheral sensory<br>neuropathy | 158 (47)                                              | 15 (5)              | 88 (52)                                             | 8 (5)               |
| Diarrhea                         | 153 (46)                                              | 16 (5)              | 62 (36)                                             | 5 (3)               |
| Headache                         | 123 (37)                                              | 5 (2)               | 61 (36)                                             | 3 (2)               |
| Vomiting                         | 123 (37)                                              | 14 (4)              | 62 (36)                                             | 3 (2)               |
| Constipation                     | 115 (34)                                              | 2 (1)               | 56 (33)                                             | 1 (1)               |
| Asthenia                         | 85 (25)                                               | 9 (3)               | 43 (25)                                             | 3 (2)               |
| Hypomagnesemia                   | 83 (25)                                               | 2 (1)               | 36 (21)                                             | 0                   |
| Dizziness                        | 70 (21)                                               | 1 (0.3)             | 30 (18)                                             | 1 (1)               |
| Pain in extremity                | 69 (21)                                               | 2 (1)               | 30 (18)                                             | 1 (0.3)             |
| Back pain                        | 40 (12)                                               | 2 (1)               | 37 (22)                                             | 8 (2)               |
| Peripheral edema                 | 68 (20)                                               | -                   | 18 (11)                                             | 1 (1)               |
| Decreased appetite               | 80 (24)                                               | 3 (1)               | 47 (28)                                             | -                   |
| Cough                            | 75 (22)                                               | -                   | 29 (17)                                             | -                   |
| Dyspnea                          | 72 (21)                                               | 3 (1)               | 34 (20)                                             | 2 (1)               |
| Arthralgia                       | 75 (22)                                               | -                   | 47 (28)                                             | 4 (2)               |
| Pyrexia                          | 67 (20)                                               | 40 (12)             | 37 (22)                                             | 13 (8)              |

paclitaxel in patients with *BRCA*-mutated, *HER2*-negative advanced breast cancer. One limitation to this data analysis is the potential effect of crossover to unblinded veliparib in the control group.

The median survival times were 32.4 months and 28.2 months for the veliparib and control groups. These durable survival times suggest the combinatorial approach is beneficial. The open-label, phase III OlympiAD study compared olaparib monotherapy with treatment of physicians' choice (TPC; excluding platinum-based chemotherapy) in patients with germline *BRCA*-mutated and *HER2*-negative metastatic breast cancer with ≤ 2 prior lines of chemotherapy for metastatic disease. Median OS with olaparib was 19.3 months vs 17.1 months with TPC (HR, 0.9; *P* = .513) at the final analysis [10]. Observed median OS was comparable for talazoparib monotherapy in the EMBRACA trial [11]. The trial populations were different, with more patients treated in second line in OlympiAD vs BROCADE3 (71% vs 19%) and exposed to prior platinum therapy (29% vs 8%). In an analysis of OlympiAD OS data for patients treated in first line, median OS was 22.6 months and 14.7 months with olaparib and TPC, respectively [12]. Moreover, crossover to PARP inhibitor was more frequent in BROCADE3 (48%) than in OlympiAD (12.4%). However, the number of long-term responders in both trials suggests PARP inhibition and platinum therapy is beneficial in this population.

The updated safety data for veliparib, including patients with long-term exposure, are consistent with those reported in the primary analysis; the addition of veliparib was generally well tolerated.

The improvement in PFS by adding veliparib to paclitaxel and carboplatin shown in the primary analysis of BROCADE3 did not translate into an OS benefit in this final analysis. The long survival times observed in both arms suggest that combination therapy with paclitaxel and carboplatin provides clinical benefit in this population.

### Role of the funding source

AbbVie funded this study and participated in the study design, research, analysis, data collection, interpretation of data, reviewing, and approval of the manuscript. All authors had access to relevant data and participated in the drafting, review, and approval of this manuscript. No

honoraria or payments were made for authorship.

### CRedit authorship contribution statement

**Ratajczak Christine:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Adamu Hephzibah:** Writing – original draft, Writing – review & editing. **Puhalla Shannon L.:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Loirat Delphine:** Writing – original draft, Writing – review & editing. **Diéras Véronique:** Conceptualization, Writing – original draft, Writing – review & editing, Investigation. **Han Hyo S.:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Girardi Vincent:** Formal analysis, Writing – original draft, Writing – review & editing. **Arun Banu K.:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Ayoub Jean-Pierre:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Wildiers Hans:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Friedlander Michael:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing.

### Declaration of Competing Interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: V. Diéras: Consulting/advisory role, Roche/Genentech, Novartis, Lilly, Pfizer, AbbVie, MSD, Daiichi Sankyo, SEAGEN, AstraZeneca, Gilead, Eisai, Medac GmbH, Menarini. Hyo S. Han: Research support (institution), AbbVie, Arvinas, Celcuity, GSK, G1 Therapeutics, Pionyr Immunotherapeutics, Quantum Leap Healthcare Collaborative, Marker, Mersana, Pfizer, Senhwa Biosciences, Zymeworks, and Department of Defense; steering committee (nonpaid), AbbVie. H. Wildiers: Consulting fees and honoraria (institution), Roche, AstraZeneca, Amgen, Lilly, Novartis, AbbVie, Vifor Pharma, Pfizer, Celldex Therapeutics, Janssen-Cilag, TRM Oncology, PUMA Biotechnology, Orion Corporation; unrestricted research grant, Roche; travel support, Roche, Pfizer. M. Friedlander: Consulting/advisory role, AstraZeneca, MSD, AbbVie, Incyclix, Novartis; speakers' bureau, AstraZeneca, MSD, GSK; honoraria, AstraZeneca, MSD, Novartis; research funding, BeiGene, AstraZeneca, Novartis. J.P. Ayoub: Research funding (institution), AbbVie, Boston Biomedical; consultancy, AstraZeneca, Eisai, Eli Lilly, Novartis, Pfizer, Puma, Roche. S.L. Puhalla: Consultant, AbbVie, MedImmune, Celldex, Puma, Pfizer, AstraZeneca, Eisai, NanoString; research funding to institution, AbbVie, Pfizer, Lilly, Novartis, Incyte, Covance-Bayer, AstraZeneca, Genentech, Medivation. D. Loirat: Honorarium/board, MSD, Gilead Sciences, AstraZeneca, Pfizer, Novartis, Roche, Lilly, Eisai, Exact Sciences; travel/congress: MSD, Gilead Sciences, AstraZeneca, Pfizer, Novartis. B.K. Arun: Research support (institution), AbbVie, PharmaMar, AstraZeneca, Invitae; steering committee (nonpaid), AbbVie. C. Ratajczak, H. Adamu, V. Girardi: AbbVie employees and may own stock.

### Data Availability

AbbVie is committed to responsible data sharing regarding the clinical trials we sponsor. This includes access to anonymized, individual and trial-level data (analysis data sets), as well as other information (eg, protocols and clinical study reports), as long as the trials are not part of an ongoing or planned regulatory submission. This includes requests for clinical trial data for unlabeled products and indications. These clinical trial data can be requested by any qualified researchers who engage in rigorous, independent scientific research, and will be provided following review and approval of a research proposal and Statistical Analysis Plan and execution of a Data Sharing Agreement. Data requests can be submitted at any time and data will be accessible for 12 months, with possible extensions considered. For more information on the

process, or to submit a request, visit the following link: <https://www.abbvie.com/our-science/clinical-trials/clinical-trials-data-and-information-sharing/data-and-information-sharing-with-qualified-researchers.html>.

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